Yiyan Liao

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EDUCATION

School of Life Sciences, Peking University Sep. 2023 – Jun. 2027 (Expected)

Bachelor of Engineering in Bioinformatics

- Ranking: 4/133; GPA: 3.75/4.00
- Selected Coursework: Linux-based Essential Bioinformatics (A); Innovative Experiments in Bioinformatics (A); Bioinformatics (A); Artificial Intelligence for Protein Design (94/100); Computational Systems Biology (93/100)

SELECTED PUBLICATIONS

†Equal contribution. *Corresponding author.

[1] **MGTbind: a comprehensive database of molecular glue ternary interactome**
Jintao Zhu†, ***Yiyan Liao***†, Haoyu Lin†, ..., Luhua Lai*, Jianfeng Pei*
Nucleic Acids Res., 54(D1), D1500–D1509, 2026. https://doi.org/10.1093/nar/gkaf1075

[2] **Benchmarking Cofolding Methods for Molecular Glue Ternary Structure Prediction**
Yiyan Liao†, Jintao Zhu†,*, Juan Xie, Luhua Lai*, Jianfeng Pei*
J. Chem. Inf. Model., 65(20), 11136–11148, 2025. https://doi.org/10.1021/acs.jcim.5c01860

[3] **DiffTernary: a Diffusion Model for Molecular Glue Ternary Complex Docking**
Yiyan Liao, Letian Lin, Jintao Zhu*, Luhua Lai*, Jianfeng Pei*
bioRxiv preprint, 2026. https://doi.org/10.64898/2026.11.09.765352

[4] **Molexar: A Unified Multimodal Molecular Foundation Model for Drug Design**
Haoyu Lin†, ***Yiyan Liao***†, Jinmei Pan, Xinliao Ling, Luhua Lai*, Jianfeng Pei*
arXiv preprint, 2026. https://doi.org/10.48550/arXiv.2606.25865

[5] **MLIPDiff: Guiding Denoising Diffusion Models for Structure-Based Drug Design via Machine-Learned Interatomic Potentials**
Yiyan Liao†, Amin Tavakoli†, Yu Zhang, Anima Anandkumar*
2nd SIMBIOCHEM Workshop at NeurIPS, 2026. https://doi.org/10.48550/arXiv.2610.33315

RESEARCH EXPERIENCE

Anima AI + Science Lab, California Institute of Technology Pasadena, CA

Summer Undergraduate Research Fellow (SURF) — Advisor: Dr. Anima Anandkumar Mar. 2026 – Present

MLIPDiff: Physics-Guided Molecular Generation (NeurIPS SIMBIOCHEM 2026; ongoing)

- Co-developed training-free, inference-time guidance for TargetDiff using quantum-trained UMA potentials and four energy terms covering protein–ligand interaction, ligand strain, steric clash, and pocket anchoring.
- Eliminated protein–ligand collisions on CrossDocked2020 while preserving local geometry and achieving the best median Vina Dock score (-7.87 kcal/mol) and a 59.6% median high-affinity fraction across baselines.
- Fine-tuned TargetDiff with DDPO using UMA-computed interaction-energy rewards, achieving similar gains. Ongoing extensions target autoregressive (Pocket2Mol) and flow-matching (FLOWR.ROOT) models.

Molecular Design Laboratory, Peking University Beijing, China

Undergraduate Researcher — Advisors: Dr. Jianfeng Pei and Dr. Luhua Lai Sep. 2024 – Present

MGTbind: a Comprehensive Database of Molecular Glue Ternary Interactome (Co-first author, Nucleic Acids Res.)

- Curated 3,093 molecular glues and 3,924 ternary interactions from 227 publications for a searchable database, including molecular structures, binary and ternary binding affinities, and bioactivity measurements.
- Integrated 317 experimentally determined and 3,607 AlphaFold 3-predicted ternary structures. Annotated physicochemical descriptors, AlphaFold confidence metrics, and protein–ligand interactions.

Benchmarking Cofolding Methods for Molecular Glue Ternary Structure Prediction (First author, J. Chem. Inf. Model.)

- Curated MG-PDB, a dataset of 221 experimental molecular glue ternary structures, and constructed MGBench from 88 complexes released after AlphaFold 3's training cutoff to evaluate generalization to unseen structures.
- Benchmarked five state-of-the-art cofolding models, including AlphaFold 3, and found that performance often reflects memorized structural homology rather than transferable atomic-level rules, especially for novel E3 ligases.

DiffTernary: A Diffusion Model for Molecular Glue Ternary Complex Docking (First author, bioRxiv preprint)

- Developed an SE(3)-equivariant diffusion model for joint protein rigid-body docking and molecular-glue torsion refinement. Built Ternary-PDB, a dataset of 76,702 experimentally determined ternary complexes, using AF3-aligned preprocessing and training cutoff.
- Achieved acceptable PPI-interface prediction rates comparable to AlphaFold 3 on MGBench (50.0% vs. 51.1%) and outperformed DeepTernary on TernaryDB MG test set (41.5% vs. 30.9%).

Molecular Design Laboratory, Peking University (continued)

Molexar: A Unified Multimodal Molecular Foundation Model for Drug Design (Co-first author, arXiv preprint)

- Co-developed the Fragment-SELFIES molecular language and a Gemma 2-based autoregressive decoder conditioned on properties, pharmacophores, sequences, and binding pockets.
- Led evaluation against 20+ baselines, achieving 100% validity in unconditional and fragment-constrained generation and recovering known actives and scaffolds on MolGenBench.
- Led candidate generation and screening for DPP4, ACO1, ACOD1, and GPR6. Compounds are in activity assays.

Peking University International Genetically Engineered Machine (iGEM) Team Beijing, China

Wet Lab Leader — Advisor: Dr. Long Qian Jan. 2024 – Oct. 2024

General Disease Detection with Protease and Aptamer (2024 iGEM Gold Medal)

- Tested thrombin-binding aptamers with SPR/EMSA, performed mutagenesis and click chemistry conjugation.
- Directed wet-lab strategy for a 15-member team and presented the final project defense at the Grand Jamboree.

PROFESSIONAL EXPERIENCE

Tencent Beijing, China

Biological Large Language Model Intern Jun. 2025 – Dec. 2025

- Developed a six-stage multi-agent system with retrieval-augmented generation and NCBI tools for automated *de novo* plasmid design, raising benchmark accuracy from 10% to 51%.

Moonshot AI Beijing, China

Large Language Model Reinforcement Learning Intern Jan. 2025 – Jun. 2025

- Built biology-domain reinforcement-learning data and ran pilot experiments for Kimi K2 on the Graduate-Level Google-Proof Q&A (GPQA) benchmark, informing the full-scale training runs.

SELECTED HONORS & AWARDS

National Scholarship for Undergraduate Students (top 4% of the cohort) 2026

Caltech Summer Undergraduate Research Fellowship ($8,110 Grant) 2026

School of Life Sciences 87 Excellent Research Scholarship, First Prize (1st out of 10 finalists) 2026

Peking University Merit Student (top 10% of the cohort) 2025, 2026

Beijing Natural Science Foundation Undergraduate Research Program ($7,000 Grant) 2025

International Genetically Engineered Machine (iGEM) 2024 Gold Medal (Wet Lab Leader) 2024

34th International Biology Olympiad (IBO) Gold Medal (3rd out of 293) 2023

ACADEMIC ACTIVITIES & SERVICE

Teaching Assistant of Biochemistry Laboratory, Peking University Sep. 2025 – Jun. 2026

- Instructed 25 students per semester in weekly biochemistry laboratory sessions.
- Established an *in silico de novo* design workflow for the GST enzyme using the protein language model ESM3.
- Awarded the Outstanding Teaching Assistant Award, School of Life Sciences, Peking University (Jun. 2026).

Student Advisor of Peking University iGEM Team 2025 Jan. 2025 – Oct. 2025

- Selected members and guided research for Team Peking 2025, which won a Global Top 10 ranking.

Member of Undergraduate Honors Program in Biology (UHPB), Peking University Sep. 2024 – Present

- Gave 4 journal club talks (1 hour each) and attended 16 more, once presenting to the senior author of the paper.
- Delivered English talks on the molecular glue computational landscape (2025) and physics-guided molecular generation (2026, Best Poster Presenter Award) at UHPB Annual Symposia, to 100+ peers and 10 professors, including international faculty.

SKILLS & OTHER INFO

- **Machine Learning Methods:** Diffusion Models, SE(3)-Equivariant Neural Networks, Autoregressive Models, Reinforcement Learning (DDPO), Supervised Fine-Tuning (SFT), Multi-Agent Systems
- **Programming & Computational Tools:** Python, PyTorch, PyTorch Geometric, ASE, RDKit, R, Bash, Git, Linux, SLURM, Schrödinger, GROMACS, OpenMM, Rosetta, AutoDock Vina, AlphaFold 3, ESM3, PyMOL, ChimeraX
- **Experimental Methods:** Molecular Cloning, Gibson Assembly, Golden Gate Assembly, Site-Directed Mutagenesis, Protein Expression and Purification, Ni-NTA Affinity Chromatography, ÄKTA FPLC, Enzyme Activity and Kinetic Assays, SPR, EMSA, SDS-PAGE, Western Blot, Co-IP, PCR/qPCR, Bacterial Culture
- **Languages:** TOEFL iBT: 103/120 (Aug. 2025), 5.0/6.0 (Apr. 2026, new scale)